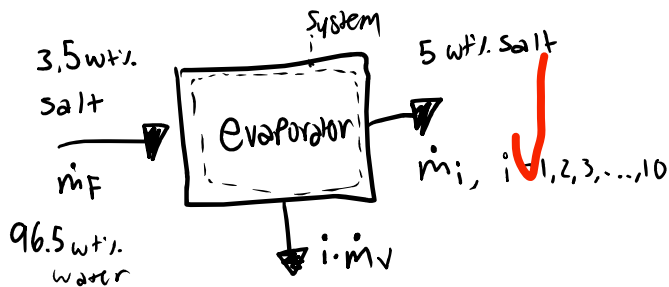


- (1) Seawater containing 3.5 wt% salt and mass flow rate $\dot{m}_F$ is fed into a series of 10 evaporators. Equal mass flow rates of pure water are vaporized in each of the 10 units and then condensed and combined to obtain a product stream of fresh water of mass flow rate $\dot{m}_W$. The brine leaving each evaporator (except for the tenth and final unit) is fed into the next evaporator in series. The brine mass flow rate leaving the first evaporator is $\dot{m}_1$, leaving the second evaporator is $\dot{m}_2$, and so on. The brine leaving the tenth evaporator contains 5 wt% salt and has a mass flow rate of $\dot{m}_{10}$.
- (a) Draw a flowchart of the process. Label all the necessary streams. Can you draw your flow diagram in such a way as not to need to show all ten units explicitly?
- (b) Write the set of equations you would solve to determine the fractional yield of fresh water from the process (kg H₂O recovered/kg H₂O in feed) and the mass fraction w_{45} of salt in the solution leaving the fourth evaporator. Each equation you write should contain no more than one previously undetermined variable. In each equation, circle the variable for which you would solve.
- (c) Solve the equations derived in part (b) for the two specified quantities. Does the problem need a basis?
- (d) Derive a general expression for the mass fraction of salt leaving the n^{th} evaporator, w_{nS} , as a function of n .

a.)



b.)

mass balance

$$\dot{m}_F = \dot{m}_i + i \cdot \dot{m}_v$$

$$\boxed{\dot{m}_i = i \cdot \dot{m}_v - \dot{m}_F}$$

Salt

$$0.035 \dot{m}_F = w_{iS} \dot{m}_i$$

$$\boxed{w_{iS} = \frac{0.035 \cdot \dot{m}_F}{\dot{m}_i}}$$

$$\dot{m}_F = \dot{m}_{10} + \dot{m}_W, \quad \dot{m}_W = 10 \dot{m}_v$$

$$\dot{m}_{10} = \frac{0.035 \dot{m}_F}{w_{10S}}$$

$$\Rightarrow \dot{m}_F = \frac{0.035 \dot{m}_F}{w_{10S}} + 10 \dot{m}_v$$

$$\Rightarrow w_{10S} (\dot{m}_F - 10 \dot{m}_v) = 0.035 \dot{m}_F$$

H₂O fractional
yield

$$\boxed{y_{\text{rel}} = \frac{10 \dot{m}_v}{0.965 \dot{m}_F}}$$

$$\dot{m}_F = \dot{m}_U + 4\dot{m}_V$$

$$\Rightarrow \dot{m}_U = \dot{m}_F - 4\dot{m}_V$$

$$w_{us} = \frac{0.035\dot{m}_F}{\dot{m}_U} = \frac{0.035\dot{m}_F}{\dot{m}_F - 4\dot{m}_V} = w_{us}$$

c.)

$$0.035\dot{m}_F = w_{10s}(\dot{m}_F - 10\dot{m}_V)$$

$$\Rightarrow 10\dot{m}_V = -\frac{0.035\dot{m}_F}{w_{10s}} + \dot{m}_F \Rightarrow$$

$$\dot{m}_V = \frac{\dot{m}_F}{10} - \frac{0.035\dot{m}_F}{10w_{10s}}$$

Sub into yield eqn

$$yield = \frac{10 \cdot \left(\frac{\dot{m}_F}{10} - \frac{0.035\dot{m}_F}{10w_{10s}} \right)}{0.965\dot{m}_F} = \left(1 - \frac{0.035}{w_{10s}} \right) \cdot \frac{1}{0.965}, w_{10s} = 0.05$$

$$yield = 0.3109 = \boxed{31.09\%}$$

$$w_{us} = \frac{0.035\dot{m}_F}{\dot{m}_F - 4\dot{m}_V} = \frac{0.035\dot{m}_F}{\dot{m}_F - 4\left(\frac{\dot{m}_F}{10} - \frac{0.035\dot{m}_F}{10w_{10s}} \right)} = \frac{\cancel{\dot{m}_F} 0.035}{\cancel{\dot{m}_F} \left(1 - 4\left(\frac{1}{10} - \frac{0.035}{10 \cdot 0.05} \right) \right)}$$

$$w_{us} = 0.0398$$

d.) $\dot{m}_n = \dot{m}_F - n \cdot \dot{m}_V$

$$w_{ns} = \frac{0.035 \cdot \dot{m}_F}{\dot{m}_F - n \dot{m}_V}$$

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(2) Biological organisms need an environment of humid air enriched in oxygen in order to grow. To provide a continuous stream of appropriate humid air, the following three input streams are fed into an evaporator unit.

- (1) Air (21 mol% O₂, the balance N₂)
- (2) Liquid water, fed at a rate of 20.0 cm³/min
- (3) Pure oxygen, with a molar flow rate one fifth of the molar flow rate of the air input.

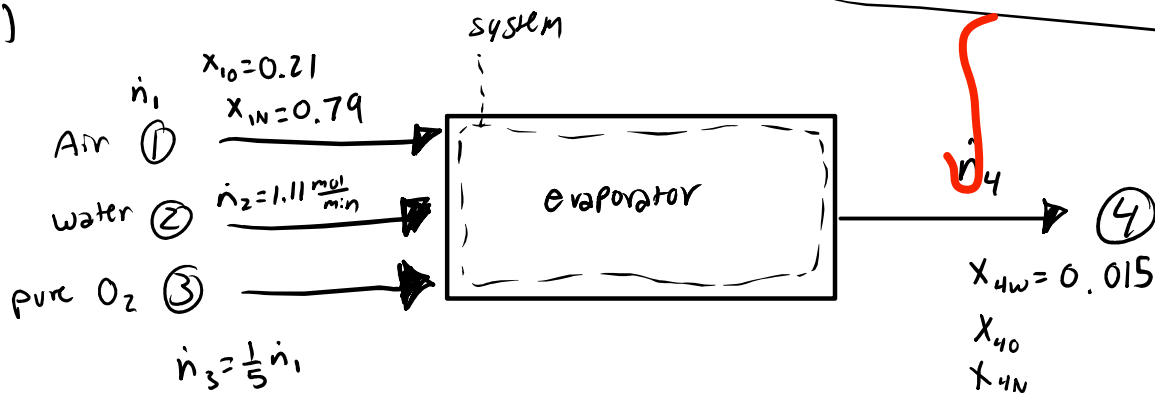
The single, gas output stream contains 1.5 mole % water.

- (a) Draw and label a flow diagram of the process. Label the total molar flow rates of the air, water, and pure oxygen input streams as $\dot{n}_1$, $\dot{n}_2$, and $\dot{n}_3$, respectively. For the exit stream label the total molar flow rate as $\dot{n}_4$ and the mole fractions of oxygen, nitrogen, and water as $x_{4,O}$, $x_{4,N}$, and $x_{4,W}$, respectively.
- (b) Do a degree of freedom analysis.
- (c) Calculate all unknown stream variables.

Data for water: $M_{H_2O} = 18.015 \text{ g/mol}$ and density 1.00 g/cm^3 .

$$\dot{V}_2 = 20 \frac{\text{cm}^3}{\text{min}} \cdot \frac{1.00 \text{ g}}{\text{cm}^3} \cdot \frac{1 \text{ mol}}{18.015 \text{ g}} = \dot{n}_2 = 1.11 \frac{\text{mol}}{\text{min}}$$

2.)



b.)

$$D_oF = (4a + 4b) - (5a + 5b + 5c + 5d)$$

$$D_oF = (7 + 0) - (2 + 2 + 0 + 3) = 7 - 7 = 0, \text{ well posed problem}$$

c.)

$$\dot{n}_2 = 1.11 \frac{\text{mol}}{\text{min}}$$

$$\dot{n}_1 + \dot{n}_2 + \dot{n}_3 = \dot{n}_4$$

$$\text{total: } \dot{n}_1 + \dot{n}_2 + \dot{n}_3 = \dot{n}_4$$

$$O_2: 0.21\dot{n}_1 + 0.2\dot{n}_3 = X_{O_2}\dot{n}_4$$

$$N_2: 0.79\dot{n}_1 = X_{N_2}\dot{n}_4$$

$$H_2O: \dot{n}_2 = 0.015\dot{n}_4 \Rightarrow \dot{n}_4 = \frac{\dot{n}_2}{0.015} = \boxed{\dot{n}_4 = 74 \frac{\text{mol}}{\text{min}}} \quad \checkmark$$

$$\dot{n}_1 + \dot{n}_2 + 0.2\dot{n}_1 = \dot{n}_4$$

$$\dot{n}_1 + 0.2\dot{n}_1 = \dot{n}_4 - \dot{n}_2$$

$$\dot{n}_1 = \frac{\dot{n}_4 - \dot{n}_2}{1.2} \Rightarrow \boxed{\dot{n}_1 = 60.74 \frac{\text{mol}}{\text{min}}} \quad \checkmark$$

$$\dot{n}_3 = 0.2\dot{n}_1 \Rightarrow \boxed{\dot{n}_3 = 12.15 \frac{\text{mol}}{\text{min}}} \quad \checkmark$$

$$\text{from } N_2 \text{ balance: } \frac{0.79\dot{n}_1}{\dot{n}_4} = X_{N_2} = \frac{0.79(60.74)}{74} = \boxed{0.648 \frac{\text{mol } N_2}{\text{mol tot}}} = X_{N_2} \quad \checkmark$$

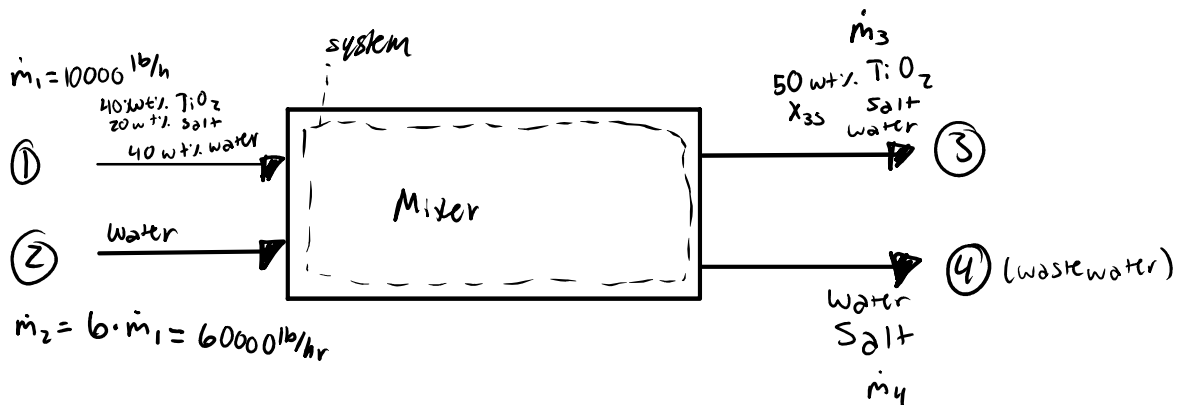
$$\text{from } O_2 \text{ balance: } \frac{0.21\dot{n}_1 + \dot{n}_3}{\dot{n}_4} = X_{O_2} = \frac{0.21(60.74) + 12.15}{74} = \boxed{X_{O_2} = 0.337 \frac{\text{mol } O_2}{\text{mol tot}}} \quad \checkmark$$

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(3) Murphy 2.66 First, solve using only two of the constraints (1. salt concentration in washed pigment and 2. optimum washwater feed rate).

P2.66 Titanium dioxide (TiO_2) is a major pigment in white paint. You are in charge of designing a washer for processing raw pigment. The washer

should be designed to handle 10,000 lb/h raw pigment (40 wt% TiO_2 , 20 wt% salt, and 40 wt% water). In the washer, the raw pigment is mixed with washwater. Pilot plant studies show that the optimum washwater feed rate is 6 lb washwater per lb raw pigment feed. 100% of the TiO_2 in the raw pigment is recovered in the washed pigment. The washed pigment product leaving the washer contains 50 wt% TiO_2 , and the salt content in the washed pigment must be no more than 100 parts salt per 1 million parts TiO_2 . The plant manager would like to dump the wastewater into a nearby river. The local water quality control regulatory agency will allow at most a discharge of 30,000 lb/h at a maximum salt content of 0.5%. Draw a flow diagram. Complete a DOF analysis and state whether this process is completely specified. If the problem is incorrectly specified, explain how you might change the design to fix the situation. Clearly explain your reasoning. (Adapted from G. V. Reklaitis, *Introduction to Material and Energy Balances*, Wiley, 1983.)



$$\text{DoF: } (4a+4b) - (5a+5b+5c+5d)$$

$$\text{DoF} = (9+0) - (2+4+0+3) = 9-9=0, \text{ well posed problem, completely specified}$$

balance Eqs.

$$\text{tot: } \dot{m}_1 + \dot{m}_2 = \dot{m}_3 + \dot{m}_4$$

$$\text{TiO}_2: 0.4\dot{m}_1 = 0.5\dot{m}_3$$

$$\text{salt: } 0.2\dot{m}_1 = x_{3s}\dot{m}_3 + x_{4s}\dot{m}_4$$

$$\text{H}_2\text{O: } 0.4\dot{m}_1 + \dot{m}_2 = x_{3w}\dot{m}_3 + x_{4w}\dot{m}_4$$

$$\dot{m}_2 = 60000 \frac{\text{lb}}{\text{hr}}$$

~~X~~ - 2 pts

from T:O₂: $\frac{0.4\dot{m}_1}{0.5} = \dot{m}_3, \{\dot{m}_1 = 10000 \frac{\text{lb}}{\text{hr}}\} \Rightarrow \boxed{\dot{m}_3 = 8000 \frac{\text{lb}}{\text{hr}}}$

from tot: $\dot{m}_1 + \dot{m}_2 - \dot{m}_3 = \dot{m}_4, \{\dot{m}_1 = 10000 \frac{\text{lb}}{\text{hr}}, \dot{m}_2 = 60000 \frac{\text{lb}}{\text{hr}}, \dot{m}_3 = 8000 \frac{\text{lb}}{\text{hr}}\}$

$\boxed{\dot{m}_4 = 62000 \frac{\text{lb}}{\text{hr}}}$

$\underline{x_{3s} = 0.005}$, from salt: $\frac{0.2\dot{m}_1 - x_{3s}\dot{m}_3}{\dot{m}_4} = x_{4s}, \{\dot{m}_1 = 10000, \dot{m}_3 = 8000, \dot{m}_4 = 62000\} (=) \frac{\text{lb}}{\text{hr}}$

$\underline{x_{4s}} = 0.0316 \rightarrow \boxed{3.16 \text{ wt}\% \text{ salt in stream 4}}$

$\downarrow$
 $1 - x_{4s} = \underline{x_{4w}} = 0.9684 \rightarrow \boxed{96.84 \text{ wt}\% \text{ water in stream 4}}$

from water:

$\frac{0.4\dot{m}_1 + \dot{m}_2 - x_{4w}\dot{m}_4}{\dot{m}_3} = x_{3w}, \{\dot{m}_1 = 10000, \dot{m}_4 = 62000, \dot{m}_3 = 8000, \dot{m}_2 = 60000, x_{4w} = 0.9684\}$

$\underline{x_{3w}} = 0.4949$

$\boxed{49.49 \text{ wt}\% \text{ water in stream 3}}$

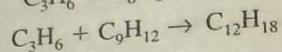
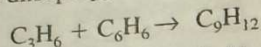
was not necessary, $1 - 0.5 - 0.005 = 0.495$
could have done $\rightarrow$ to get same result

explanation?
— 6 pts

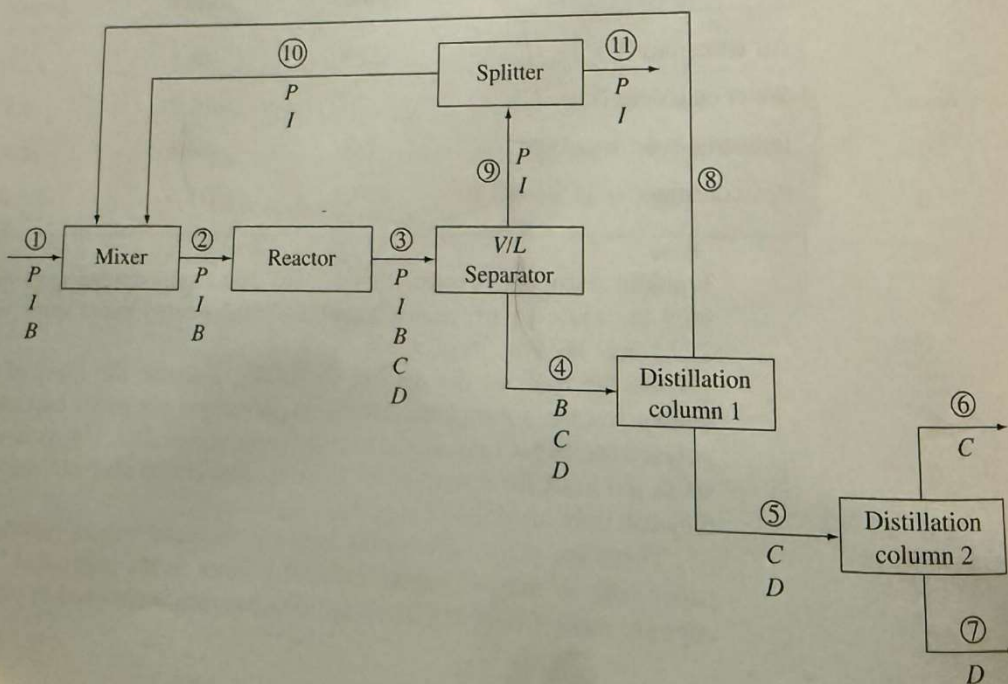
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- (4) Murphy 2.69 Although the problem statement says you do not have to solve the equations, you should try to solve them anyway. Think about strategies to find a solution and how you might use a DoF to guide you. But do not hand in your attempted solution for grading. The solution to the balance equations is a challenging problem. It should be your goal to solve this problem completely in the next few weeks.

assumed in the Flamm... counted as solid wastes. Which approach do you...
P2.69 Refer to the block flow diagram below. A 95 mol% propylene P (C_3H_6)/5 mol% propane I (C_3H_8) feed is mixed with benzene B (C_6H_6) feed at a molar ratio of 1.2:1 propylene:benzene ($P:B$). These fresh feeds are mixed with recycled streams, then fed to a reactor. The stream entering the reactor contains 10 mol% propane. Two reactions occur simultaneously in the reactor, producing the desired product cumene C (C_9H_{12}) and an undesired byproduct diisopropylbenzene D ($C_{12}H_{18}$):



Under the reaction conditions, propane I is an inert. 80% of the propylene and 72% of the benzene fed to the reactor are converted to products. The reactor effluent is cooled and sent to a separator, where a vapor stream containing propylene and propane is taken off the top and a liquid stream containing benzene, cumene, and diisopropylbenzene is taken off the bottom. A fraction of the vapor stream leaving the separator is purged and the remainder is recycled to be mixed with the incoming feed stream. The liquid stream leaving the reactor is sent to a series of two distillation columns, where benzene is recovered and recycled, and cumene and diisopropylbenzene are separated and sent to storage tanks. The cumene production rate is 25 gmol/s.



Complete a DOF analysis on the process. On the basis of the DOF analysis, immediately write down the set of variables and equations that you would use to calculate flows of all streams. You do not have to solve the equations.

▲ Mixer:

SL:

$$\frac{P_1}{I_1} = \frac{95}{5}, \quad \frac{P_1}{B_1} = \frac{1.2}{1}$$

$$SZ: 0.1 = \frac{I_2}{P_2 + I_2 + B_2}$$

$$P: P_{10} + P_1 = P_2$$

$$I: I_{10} + I_1 = I_2$$

$$B: I_1 + I_8 = I_2$$

▲ reactor:

$$\frac{P_{cons1}}{C_{gen1}} = 1, \quad \frac{B_{cons1}}{C_{gen1}} = 1, \quad \frac{P_{cons1}}{B_{cons1}} = 1$$

$$\frac{P_{cons2}}{D_{geno}} = 1, \quad \frac{C_{cons2}}{D_{gen2}} = 1, \quad \frac{P_{cons2}}{C_{cons2}} = 1$$

$$P: 0.8P_2 - P_3 - P_{cons1} - P_{cons2} = 0$$

$$B: 0.72B_2 - B_3 - B_{cons2} = 0$$

$$I: I_1 = I_3$$

$$C: -C_3 + C_{gen2} - C_{cons2} = 0$$

$$D: D_{gen2} - D_3 = 0$$

▲ Separator

$$P: P_3 - P_4 = 0$$

$$B: B_2 - B_4 = 0$$

$$I: I_3 - I_4 = 0$$

$$C: C_3 - C_4 = 0$$

$$D: D_3 - D_4 = 0$$

A Splitter

$$P: P_9 - P_{10} - P_{11} = 0$$

$$I: I_9 - I_{10} - I_{11} = 0$$

A Distillation Column 1

$$B: B_4 - B_8 = 0$$

$$C: C_4 - C_5 = 0$$

$$D: D_4 - D_5 = 0$$

A Distillation Column 2

$$C: C_5 - C_6 = 0$$

$$D: D_5 - D_7 = 0$$

$$\text{DoF} = (4a + 4b) - (5a + 5b + 5c + 5d)$$

$$\text{DoF} = (25 + 2) - (1 + 4 + 2 + 20) = 27 - 27 = \underline{0}$$